Access-Constrained Clarification for Efficient and Safe Service in Text-to-SQL (ACCESS) Supplementary Analyses

This supplement provides the evaluation details behind ACCESS's main results. It uses the same recorded SQL, request populations, pipeline names, and order-aware metrics as the main paper. The main comparisons use the frozen 2,800-request cohort; Section S16 examines completion across the full planned population.

The benchmark-allowed set A contains 1,402 requests. The benchmark-denied requests comprise 80 *recoverable denied requests* in P, each with a nonempty authorized output reference, and 1,318 reference-negative requests in Z. These sets are disjoint. Output fulfillment is evaluated on $E = A \cup P$, and safe handling on $\mathcal{I}_{\text{den}} = P \cup Z$. The notation $|S|$ denotes the number of requests in a set S. The terms *recoverable denied* and *reference-positive benchmark-denied* refer to the same fixed set P.

BSS is the harmonic mean of SafeEX and SHR. Scores are defined on $[0, 1]$ and reported as percentages; differences are in percentage points (pp). Unless explicitly labeled as bootstrap intervals or variation across models, ranges represent lower and upper scores supported by the available evaluation evidence. Rescoring reuses recorded SQL without new model inference.

Each analysis supports a specific part of the main paper. Sections S1 and S7 explain reference construction and evaluation. Sections S2–S4 and S15 expand the main comparison in Section V-A (Tables II–III); S10–S12 expand the ablations in Section V-B (Table IV); S5–S6 explain the outcomes in Section V-D (Table V); S9 and S14 support interaction costs in Section V-E (Table VI); and S8 and S13 support the reference and exact-output checks in Section V-F (Tables VIII and VII, respectively). Sections S16–S17 document the cohort and specificity construction in Section IV-A.

S1 Fixed Reference Construction and Validation

The benchmark labels define the request-level evaluation groups for every pipeline. Output references are constructed independently of candidate SQL and scores. Each reference witness must pass permission checks and reproduce the required result tuples, duplicate multiplicity, and final ordering. Candidate SQL is evaluated separately under the same role policy.

The construction examines all 2,234 benchmark-denied output units and identifies 80 reference-positive requests. An output unit is an output position in a request's Gold result. Two human reviewers independently examined all 1,318 reference-negative requests and found no additional authorized outputs preserving the original requirements. The reference construction retains essential dependencies, complete result tuples, and required final ordering.

SQL witness checks confirm all 80 references on their benchmark databases. Two human reviewers independently assessed the references; their consolidated assessment accepted all 80 (100%). Within this set, 47 requests have structurally supported partial-output references. Section S8 separates subsets by retained output positions, operator structure, and output content.

Some references retain only grouping labels; these occur in `pets_1/258` and `singer/44`. In contrast, 12 cases in four source-question clusters would require changing the Gold ordering and are assigned to the reference-negative set Z. Section S8 explains the output and operator diagnostics, and Section S9 records the human-review procedure.

S2 Safe Handling Combines Refusal and Authorized Delivery

Refinement returns a full handoff, partial handoff, refusal, or unresolved outcome, as defined in Section II of the main paper. Handoffs proceed to SQL generation; refusal ends service; unresolved indicates that refinement has not reached a justified terminal decision. For request i, the indicator ref_i records a terminal refusal, and safe_i records delivery of a read-only SQL query that passes native authorization. The indicators are mutually exclusive. Authorized delivery earns safe-handling credit even if execution or output matching later fails. On recoverable denied requests P, whole-request refusal earns SHR credit and zero SafeEX credit. The component rates are

$$\text{CRR}_Z = \frac{\sum_{i \in Z} \text{ref}_i}{|Z|}, \qquad \text{SDR}_P = \frac{\sum_{i \in P} \text{safe}_i}{|P|}, \qquad \text{SHR}_P = \frac{\sum_{i \in P} (\text{ref}_i + \text{safe}_i)}{|P|}.$$

Here, CRR is the correct-refusal rate on Z, SDR is the authorized SQL-delivery rate on P, and SHR_P includes both refusals and authorized deliveries. For any nonempty subset $\Omega \subseteq E$, Full_Ω is the fraction with fulfillment score $c_i = 1$: all target columns jointly match complete reference rows, duplicate counts, and required order, and the entire query is authorized. Full permits extra authorized candidate columns; Section S13 evaluates them separately. Matching empty results retains $[0, 1]$ bounds when column correspondence is uncertain, unless another confirmed failure establishes zero. If only one result is empty, the row mismatch scores zero.

Table S1: Safe-handling decomposition

Model	Pipeline	CRR_Z	Ref_P	SDR_P	SHR_P	Full_P
DeepSeek Flash	Direct	63.58	38.75	40.00	78.75	7.50
DeepSeek Flash	Serial	58.35	55.00	26.25	81.25	5.00
DeepSeek Flash	Policy	28.45	18.75	36.25	55.00	16.25
DeepSeek Flash	ACCESS	88.01	18.75	73.75	92.50	48.75
Qwen2.5-14B	Direct	24.81	20.00	33.75	53.75	18.75
Qwen2.5-14B	Serial	28.15	30.00	12.50	42.50	2.50
Qwen2.5-14B	Policy	28.53	18.75	22.50	41.25	8.75
Qwen2.5-14B	ACCESS	88.01	18.75	61.25	80.00	40.00
Arctic R1-7B	Direct	0.08	0.00	23.75	23.75	3.75
Arctic R1-7B	Serial	0.46	3.75	8.75	12.50	0.00
Arctic R1-7B	Policy	28.45	18.75	15.00	33.75	3.75
Arctic R1-7B	ACCESS	88.01	18.75	51.25	70.00	33.75
Qwen2.5-7B	Direct	38.85	22.50	20.00	42.50	7.50
Qwen2.5-7B	Serial	51.59	43.75	8.75	52.50	1.25
Qwen2.5-7B	Policy	28.45	18.75	10.00	28.75	2.50
Qwen2.5-7B	ACCESS	88.01	18.75	73.75	92.50	46.25
Qwen3.8-27B	Direct	77.39	58.75	25.00	83.75	3.75
Qwen3.8-27B	Serial	62.97	53.75	25.00	78.75	3.75
Qwen3.8-27B	Policy	28.45	18.75	25.00	43.75	10.00
Qwen3.8-27B	ACCESS	88.01	18.75	73.75	92.50	48.75

Note. **ACCESS combines correct refusal with more safe deliveries on recoverable denied requests.** Five-model denied-handling decomposition. Ref_P is the refusal rate on P. $\text{Ref}_P + \text{SDR}_P = \text{SHR}_P$; Full_P additionally requires complete correct results.

Table S1 shows ACCESS's highest correct-refusal rate on Z and safe-delivery rate on recoverable denied requests (P) within every model. Its frozen frontend supplies 1,160 refusals on Z and 15 on P across models; safe SQL counts on P are 59, 49, 41, 59, and 59, while complete recoveries number 39, 32, 27, 37, and 39. Thus the workflow combines rejection of unserviceable requests with authorized partial deliveries, but useful output still depends on SQL realization. The gap between safe and complete deliveries measures that downstream distinction under the same frontend decisions.

S3 Recovery and Handling Make Distinct Contributions

Refuse-All provides a calibration: its zero output fulfillment yields zero BSS despite perfect SHR. To explain the gains of the evaluated pipelines, write $\mathcal{H}(x, y) = 2xy/(x + y)$ for positive $x + y$, and $\mathcal{H}(0, 0) = 0$. Let x denote

SafeEX and y denote SHR; subscripts 0 and 1 identify a comparator and ACCESS, respectively. We use confirmed lower SafeEX endpoints and the resolved SHR values. The two-component attribution averages the orders in which recovery and handling are changed:

$$C_{\text{rec}} = \tfrac{1}{2}[\mathcal{H}(x_1, y_0) - \mathcal{H}(x_0, y_0) + \mathcal{H}(x_1, y_1) - \mathcal{H}(x_0, y_1)],$$
$$C_{\text{hand}} = \tfrac{1}{2}[\mathcal{H}(x_0, y_1) - \mathcal{H}(x_0, y_0) + \mathcal{H}(x_1, y_1) - \mathcal{H}(x_1, y_0)].$$

The intermediate terms cancel, giving $C_{\text{rec}} + C_{\text{hand}} = \mathcal{H}(x_1, y_1) - \mathcal{H}(x_0, y_0)$. Each row therefore divides the observed BSS gain into contributions from output fulfillment and safe handling. This is an accounting of the score difference; the module-removal experiments appear in Sections S10–S12. Contributions are computed before rounding, so displayed parts can differ from the displayed total by 0.01 point.

Table S2: BSS gain decomposition

Model	Versus	Recovery	Handling	Total
DeepSeek Flash	Direct	+17.35	+6.33	+23.68
DeepSeek Flash	Serial	-1.62	+10.65	+9.04
DeepSeek Flash	Policy	-1.92	+29.59	+27.67
Qwen2.5-14B	Direct	+12.85	+19.70	+32.55
Qwen2.5-14B	Serial	-2.38	+24.88	+22.50
Qwen2.5-14B	Policy	-2.14	+24.54	+22.40
Arctic R1-7B	Direct	+8.86	+47.68	+56.53
Arctic R1-7B	Serial	-2.31	+59.33	+57.01
Arctic R1-7B	Policy	-2.44	+25.38	+22.94
Qwen2.5-7B	Direct	+22.70	+12.59	+35.29
Qwen2.5-7B	Serial	+11.27	+10.09	+21.37
Qwen2.5-7B	Policy	+2.37	+25.07	+27.44
Qwen3.8-27B	Direct	+23.58	+2.19	+25.77
Qwen3.8-27B	Serial	-1.97	+8.39	+6.41
Qwen3.8-27B	Policy	-1.06	+28.51	+27.44

Note. **Safe handling drives most gains over interactive baselines.** BSS attribution at confirmed endpoints, ACCESS minus comparator (pp).

Population contributions remain additive within each component, where Δ means ACCESS minus the comparator:

$$\Delta\,\text{SafeEX} = \frac{|A|}{|E|}\Delta\,\text{SafeEX}_A + \frac{|P|}{|E|}\Delta\,\text{SafeEX}_P, \qquad \Delta\,\text{SHR} = \frac{|Z|}{|\mathcal{I}_{\text{den}}|}\Delta\,\text{CRR}_Z + \frac{|P|}{|\mathcal{I}_{\text{den}}|}\Delta\,\text{SHR}_P\,.$$

For DeepSeek versus Serial, the two recovery contributions are -4.80 and $+2.30$ points; the two handling contributions are $+27.97$ and $+0.64$. The harmonic score transforms these component changes into -1.62 recovery and $+10.65$ handling points. Consequently, the population proportions weight SafeEX and SHR internally, rather than serving as fixed BSS weights. Table S2 therefore attributes the DeepSeek advantage over Serial to safer handling, chiefly correct refusal on Z; the separate recovery comparisons establish the partial-service benefit.

For the disjoint-population check, use $B_{AZ} = \mathcal{H}(\text{SafeEX}_A, \text{CRR}_Z)$, excluding P from both components. ACCESS's lower endpoint exceeds each comparator's upper endpoint for every model. Section S15 reports these margins and varies the relative emphasis using weighted harmonic aggregation.

S4 Partial-Recovery Gains Survive Paired Testing

We resample the 28 source-question clusters with replacement within each data source, keeping each stratum's cluster count fixed. All associated role/specificity variants and pipeline pairs remain together. Each of 10,000 replicates (seed 20261106) recomputes the denominator after cluster resampling. The intervals below concern SafeEX_P differences. All scoring evidence on recoverable denied requests (P) is resolved; interval width here represents cluster resampling rather than unresolved permissions.

Table S3 gives positive adjusted lower endpoints in all 15 comparisons on the fixed 28-cluster population. The higher recovery on recoverable denied requests (P) therefore persists after accounting for paired source-question sampling and multiple comparisons.

Table S3: Recovery on recoverable denied requests

Model	Versus	Difference	95% interval	Adjusted interval
DeepSeek Flash	Direct	40.94	[26.20, 54.69]	[19.34, 61.77]
DeepSeek Flash	Serial	42.60	[28.95, 55.21]	[22.31, 61.89]
DeepSeek Flash	Policy	32.19	[21.13, 42.95]	[16.67, 47.62]
Qwen2.5-14B	Direct	22.19	[9.45, 36.36]	[3.63, 43.23]
Qwen2.5-14B	Serial	38.12	[24.70, 51.88]	[19.39, 59.30]
Qwen2.5-14B	Policy	31.25	[20.00, 42.11]	[14.63, 46.84]
Arctic R1-7B	Direct	29.58	[16.87, 41.88]	[10.84, 46.37]
Arctic R1-7B	Serial	33.33	[22.08, 43.90]	[17.08, 48.32]
Arctic R1-7B	Policy	30.00	[19.48, 40.51]	[14.75, 44.87]
Qwen2.5-7B	Direct	38.33	[22.92, 53.16]	[15.83, 60.78]
Qwen2.5-7B	Serial	45.00	[29.76, 59.76]	[22.78, 67.07]
Qwen2.5-7B	Policy	42.08	[26.92, 56.79]	[19.92, 63.89]
Qwen3.8-27B	Direct	43.85	[29.49, 57.38]	[22.37, 64.22]
Qwen3.8-27B	Serial	42.60	[28.85, 55.41]	[21.95, 62.30]
Qwen3.8-27B	Policy	36.25	[22.76, 49.38]	[17.48, 56.16]

Note. **All 15 adjusted partial-recovery gains remain positive.** SafeEX_P improvement in percentage points. Adjusted intervals use percentile tails $0.05/(2 \times 15)$ for the family of 15 paired comparisons.

The overall analysis resamples 476 source-question clusters within data source and recomputes SafeEX and SHR on their eligible populations before forming BSS. For unresolved recovery evidence, lower and upper difference endpoints are resampled separately; percentile envelopes include sampling and evidence uncertainty without midpoint imputation. These intervals measure source-question uncertainty conditional on the frozen references and recorded runs.

S5 Benchmark-Allowed Outcome Assignment

Overall output fulfillment depends mainly on benchmark-allowed requests. SafeEX averages fulfillment across both benchmark-allowed requests A and reference-positive benchmark-denied requests P:

$$\text{SafeEX} = \frac{|A|}{|E|}\,\text{SafeEX}_A + \frac{|P|}{|E|}\,\text{SafeEX}_P\,.$$

Requests in A account for 94.60% of the output-evaluation cohort E, whereas P accounts for only 5.40%. For DeepSeek Flash, comparing the confirmed lower endpoints, ACCESS improves SafeEX_P over Policy by 32.19 percentage points (48.75% versus 16.56%), but its SafeEX_A is 5.94 points lower (54.60% versus 60.54%). Because SafeEX weights requests equally, the improvement on the smaller subset P does not offset the decrease on A. ACCESS therefore fulfills more authorized output on benchmark-denied requests while achieving lower overall SafeEX than Policy.

Each benchmark-allowed request is assigned to the first applicable category: recorded refusal; inconclusive score (different lower and upper bounds); complete output target; partial columns with full rows; absent SQL without refusal; unauthorized SQL; execution error; row-count mismatch; a positive unordered match lost to ordering; or another output mismatch. This precedence makes the categories disjoint and retains all 1,402 requests per pipeline. Execution errors include parsing, planning, and runtime failures. Row-count mismatches include duplicate rows; equal row counts can still fail tuple-value or ordering checks. Table V of the main paper uses the same categories.

Table S4 records 732 complete output targets for ACCESS, versus 811 for Serial and 824 for Policy with DeepSeek Flash. These are confirmed Full_A successes; SafeEX_A additionally credits partial columns with complete matching rows.

With DeepSeek Flash, the gap relative to Policy combines more ACCESS refusals and row-count mismatches with similar execution-error counts. Matching ACCESS's pruning records by request ID identifies 201 benchmark-allowed cases: 138 refusals, three complete recoveries, 13 partial-column recoveries, and 47 zero-credit deliveries. Their confirmed fulfillment scores sum to $3+7.8333 = 10.8333$. These records locate the lost output among requests affected by pruning; assessing individual removal decisions requires their dialogue and dependency evidence.

Table S4: Outcomes on benchmark-allowed requests

Model	Pipeline	C	Part	Ref	None	Unauth	Err	Rows	Ord	Other	Unc
DeepSeek Flash	Direct	450	62	0	37	56	314	302	21	146	14
DeepSeek Flash	Serial	811	49	0	26	76	86	160	10	142	42
DeepSeek Flash	Policy	824	48	29	0	99	73	153	12	116	48
DeepSeek Flash	ACCESS	732	63	138	0	58	72	188	14	123	14
Qwen2.5-14B	Direct	347	43	0	22	53	482	307	8	126	14
Qwen2.5-14B	Serial	701	23	0	27	72	287	146	2	122	22
Qwen2.5-14B	Policy	687	25	29	0	86	285	158	3	105	24
Qwen2.5-14B	ACCESS	601	37	138	8	66	251	173	6	104	18
Arctic R1-7B	Direct	382	40	0	16	113	305	337	11	180	18
Arctic R1-7B	Serial	734	25	0	17	89	255	144	4	111	23
Arctic R1-7B	Policy	716	21	29	12	85	230	167	1	119	22
Arctic R1-7B	ACCESS	626	26	138	20	40	210	196	3	124	19
Qwen2.5-7B	Direct	277	31	0	64	68	561	277	9	101	14
Qwen2.5-7B	Serial	480	22	0	341	65	300	113	2	68	11
Qwen2.5-7B	Policy	635	13	29	0	103	377	124	4	95	22
Qwen2.5-7B	ACCESS	647	39	138	4	70	71	183	5	131	114
Qwen3.8-27B	Direct	367	47	0	36	58	531	215	14	119	15
Qwen3.8-27B	Serial	788	39	0	38	74	142	151	5	140	25
Qwen3.8-27B	Policy	774	33	29	3	85	182	154	7	112	23
Qwen3.8-27B	ACCESS	702	58	138	1	45	71	190	12	126	59

Note. **Refusal and row mismatch distinguish benchmark-allowed outcomes.** Disjoint benchmark-allowed outcomes. C: complete output target; Part: partial columns with full rows; Ref: refusal; None: absent SQL without refusal; Unauth: unauthorized SQL; Err: execution error; Rows: row-count mismatch; Ord: positive match lost to ordering; Other: other output mismatch; Unc: inconclusive scoring evidence. Each row sums to 1,402.

Successful partial service requires both a serviceable residual request and SQL that produces its required output.

Refusal reduces completion on benchmark-allowed requests. On the fixed 1,402 requests in A, ACCESS records 138 terminal refusals (9.84%) and Policy records 29 (2.07%); Direct and Serial record none. Counts are identical across models because each pipeline replays the same frontend outcomes. The denominator is the full benchmark-allowed set used in the main comparison. These refusals account for part of the output-fulfillment gap.

S6 Score Uncertainty Is Confined to Benchmark-Allowed Requests

All scores on recoverable denied requests are resolved. Table S5 separates the remaining benchmark-allowed uncertainty into authorization evidence, empty-result column correspondence, and execution checks awaiting completion.

Authorization evidence for ten ACCESS candidates on recoverable denied requests is reconciled with the handling evaluation: eight use Qwen2.5-7B and two use Qwen3.8-27B. The match requires identical candidate ID, request ID, source cluster, and SQL SHA256. Each candidate has a full-row score of one conditional on authorization; ordered cases also have a recorded ordered match of one. Confirmed authorization therefore resolves their intervals from $[0, 1]$ to $[1, 1]$. SQL, policies, references, and outputs remain unchanged. The resolved SafeEX$_P$ values are 46.25% and 48.75%, respectively, and the main paper's overall scores and bootstrap analyses use these resolved records.

Among ACCESS's remaining records, authorization uncertainty occurs in LiveSQLBench: 3 cases with Qwen2.5-14B, 100 with Qwen2.5-7B, and 45 with Qwen3.8-27B. Empty-result correspondence is uncertain for 14, 15, 18, 14, and 14 Spider cases in model order; Arctic also has one BIRD execution check awaiting completion. Empty results can agree without identifying which columns correspond. These distinct evidence gaps explain the bounds on benchmark-allowed recovery; the recoverable-denied comparison is fully resolved.

S7 Evaluation Implementation and Comparison Scope

Complete tuples must preserve the required order. Candidate and Gold columns are matched injectively by their observed values. Selected columns must jointly reproduce every reference row and duplicate, without extra

Table S5: Unresolved evaluation evidence

Model	Pipeline	Authorization	Empty reference	Execution pending	Total
DeepSeek Flash	Direct	2	12	0	14
DeepSeek Flash	Serial	19	19	4	42
DeepSeek Flash	Policy	27	18	3	48
DeepSeek Flash	ACCESS	0	14	0	14
Qwen2.5-14B	Direct	0	12	2	14
Qwen2.5-14B	Serial	3	18	1	22
Qwen2.5-14B	Policy	6	16	2	24
Qwen2.5-14B	ACCESS	3	15	0	18
Arctic R1-7B	Direct	1	17	0	18
Arctic R1-7B	Serial	2	20	1	23
Arctic R1-7B	Policy	0	21	1	22
Arctic R1-7B	ACCESS	0	18	1	19
Qwen2.5-7B	Direct	0	12	2	14
Qwen2.5-7B	Serial	3	8	0	11
Qwen2.5-7B	Policy	5	17	0	22
Qwen2.5-7B	ACCESS	100	14	0	114
Qwen3.8-27B	Direct	1	14	0	15
Qwen3.8-27B	Serial	8	17	0	25
Qwen3.8-27B	Policy	5	18	0	23
Qwen3.8-27B	ACCESS	45	14	0	59

Note. **All remaining score uncertainty is confined to A.** Remaining unresolved SafeEX records after reconciliation. All are benchmark-allowed; P has no unresolved scores.

projected rows. Native `DENSE_RANK` over the Gold sort expressions defines equal-key tie blocks. Rows may permute within a block, not across blocks. Only the final outermost `ORDER BY` constrains output order; an inner or window order alone does not. The Gold execution fixes `LIMIT/OFFSET` membership even at a tied boundary. Numeric values are normalized to eight decimal places. Authorization covers all referenced expressions, including hidden filtering, joining, and ordering dependencies.

Exact additionally requires every candidate column to match. Full requires every target column to match, while permitting additional authorized columns. Exact in Section S13 adds the native result-width condition, so every candidate column participates in the full-reference correspondence. Its native-width metadata is separate from the scorer's list of matched column indices.

Common inputs support comparisons of policy placement. The Serial/Policy frontends use DeepSeek Flash with temperature 0, a 512-token response cap, a maximum of 30 turns and 80 calls, and the same requester-simulator reference-intent protocol. Their conversational backbone adapts BIRD-INTERACT commit `451fe2c`; the target contract is an explicit natural-language request. They receive the frozen schema without external knowledge or execution feedback. Policy adds permissions and prompt-level pruning/refusal during refinement; Serial supplies permissions only at SQL generation. Direct + RBAC supplies the initial request and permissions directly to the SQL model without dialogue. Policy and ACCESS use policy-free SQL-generation prompts after permission-aware refinement. The DeepSeek Flash SQL setting is temperature 0, a 4,096-token cap, no demonstrations, and disabled thinking. All four pipelines use the same role policy for final SQL authorization. Thus the comparison shares evaluation permissions while placing policy information at the stages defined by each pipeline.

Within each pipeline, five-model evaluation replays that pipeline's same frozen frontend outcomes and refined requests. Every main-table arm shares request IDs, Gold queries, role policies, fixed output targets, and evaluation rules. Request-level frontend counters cover all three interactive pipelines. S9 specifies their record linkage; S10 specifies variant recovery and S14 reconciles the available SQL-stage usage.

Executable checks and human review establish reference quality. The evaluation combines executable reference checks, source lineage, and independent human review of texts and references. The running example in the paper illustrates the design; quantitative comparisons use the recorded candidate and trace identities.

S8 Recovery Gains Persist across Reference Subsets

Four diagnostic subsets separate reference properties. Let G_i be all output-column positions in the full Gold query. The partial-projection subset $P_{\text{part}} = \{i \in P : L_i \subsetneq G_i\}$ contains 53 requests in 18 source clusters: each reference retains a nonempty proper subset of Gold output columns. Within P_{part}, P_G contains references beyond grouping-label-only outputs (47 instances, 16 clusters), and P_S contains structurally supported partial-output witnesses (47 instances, 16 clusters). $P_{SG} = P_S \cap P_G$ contains 41 instances in 14 clusters. The two 47-instance sets test different reference properties: output content and operator structure; all pipelines use identical IDs within each diagnostic. These subsets describe output-reference properties.

For each subset, we retain the ordered, full-row column scores and perform 10,000 paired, data-source-stratified source-cluster bootstrap replicates (seed 20261106). Each replicate recomputes its request denominator. Tables S6–S9 report scores and ordinary 95% intervals for ACCESS minus each comparator. Family-adjusted intervals use 15 comparisons within each subset or 60 across all four subsets. The main paper's Table VIII uses the 60-comparison adjustment. All intervals describe source-question sampling with the recorded runs and references held fixed.

Table S6: Recovery on partial-projection references

Model	Direct	Serial	Policy	ACCESS	vs. Direct	vs. Serial	vs. Policy
DeepSeek Flash	5.66	0.00	7.55	52.83	[27.8, 65.4]	[35.2, 69.4]	[29.6, 60.8]
Qwen2.5-14B	28.30	1.89	3.77	50.94	[7.4, 39.6]	[31.5, 65.4]	[29.6, 63.5]
Arctic R1-7B	5.66	0.00	5.66	50.94	[25.9, 64.2]	[34.0, 66.7]	[29.6, 61.1]
Qwen2.5-7B	9.43	0.00	3.77	52.83	[26.4, 59.6]	[35.2, 69.4]	[32.1, 65.4]
Qwen3.8-27B	0.00	0.00	9.43	52.83	[35.2, 69.4]	[35.2, 69.4]	[27.8, 59.3]

Note. **ACCESS leads recovery on partial-output references.** P_{part}: proper subsets of Gold output columns, $N = 53$. SafeEX scores (%) and ACCESS-minus-comparator 95% intervals (pp).

Table S7: Recovery without grouping-only outputs

Model	Direct	Serial	Policy	ACCESS	vs. Direct	vs. Serial	vs. Policy
DeepSeek Flash	6.38	0.00	8.51	59.57	[33.3, 70.2]	[43.5, 73.9]	[35.4, 66.0]
Qwen2.5-14B	31.91	2.13	4.26	57.45	[8.3, 43.5]	[38.3, 70.8]	[37.5, 68.1]
Arctic R1-7B	6.38	0.00	6.38	57.45	[31.2, 68.8]	[41.7, 71.7]	[35.4, 66.0]
Qwen2.5-7B	10.64	0.00	4.26	59.57	[31.9, 64.6]	[43.5, 73.9]	[39.6, 70.2]
Qwen3.8-27B	0.00	0.00	10.64	59.57	[43.5, 73.9]	[43.5, 73.9]	[33.3, 63.8]

Note. **ACCESS's recovery lead persists beyond grouping-label outputs.** P_G: grouping-label cases excluded, $N = 47$. SafeEX scores (%) and ACCESS-minus-comparator 95% intervals (pp).

Table S8: Recovery on structurally supported targets

Model	Direct	Serial	Policy	ACCESS	vs. Direct	vs. Serial	vs. Policy
DeepSeek Flash	6.38	0.00	8.51	55.32	[27.7, 68.1]	[37.5, 72.3]	[30.4, 63.0]
Qwen2.5-14B	29.79	2.13	4.26	51.06	[8.3, 35.6]	[31.2, 66.0]	[30.4, 63.0]
Arctic R1-7B	6.38	0.00	6.38	53.19	[25.5, 66.7]	[36.2, 69.6]	[31.1, 63.0]
Qwen2.5-7B	10.64	0.00	4.26	55.32	[27.1, 62.2]	[37.5, 72.3]	[33.3, 68.1]
Qwen3.8-27B	0.00	0.00	8.51	55.32	[37.5, 72.3]	[37.5, 72.3]	[30.4, 63.0]

Note. **ACCESS leads recovery on structurally supported witnesses.** P_S: structurally supported witnesses, $N = 47$. SafeEX scores (%) and ACCESS-minus-comparator 95% intervals (pp).

Tables S6–S9 show ACCESS with the highest recovery score in every model under each reference restriction. All scores in these subsets are resolved. On partial-projection references, ACCESS scores 50.94–52.83% across models. The 58.54–63.41% range under both structural and grouping exclusions likewise describes variation across models, not unresolved evidence. The recovery advantage therefore persists for strict partial-output targets and beyond grouping-label-only outputs. These diagnostics retain candidate SQL and vary only the reference category; Table S8's structurally supported witnesses additionally preserve duplicate behavior through the retained query structure.

Table S9: Recovery with both reference restrictions

Model	Direct	Serial	Policy	ACCESS	vs. Direct	vs. Serial	vs. Policy
DeepSeek Flash	7.32	0.00	9.76	63.41	[35.0, 73.8]	[47.6, 76.9]	[38.1, 68.3]
Qwen2.5-14B	34.15	2.44	4.88	58.54	[9.5, 40.0]	[40.5, 70.7]	[39.0, 67.5]
Arctic R1-7B	7.32	0.00	7.32	60.98	[31.7, 72.5]	[45.2, 73.8]	[38.1, 68.3]
Qwen2.5-7B	12.20	0.00	4.88	63.41	[34.1, 67.5]	[47.6, 76.9]	[42.9, 73.2]
Qwen3.8-27B	0.00	0.00	9.76	63.41	[47.6, 76.9]	[47.6, 76.9]	[38.1, 68.3]

Note. **The recovery lead persists under both reference exclusions.** P_{SG}: both exclusions, $N = 41$. SafeEX scores (%) and ACCESS-minus-comparator 95% intervals (pp).

Higher-replicate checks preserve the structural-subset gains. The tail-stability check cited in Section V-F of the main paper repeats the paired, data-source-stratified cluster bootstrap with 200,000 replicates for each of three seeds: 20261106, 20261107, and 20261108. It uses the same four reference subsets and the same 60-contrast family adjustment, with percentile tails $0.05/(2 \times 60)$ and $1 - 0.05/(2 \times 60)$. For all five ACCESS-minus-Policy contrasts on P_S summarized in the main paper's Table VIII, the adjusted lower endpoint remains positive in every run. These additional runs check the stability of the reported comparison directions; the main table retains the original 10,000-replicate intervals.

S9 Frontend Configuration, State Contracts, and Interaction Records

Record identities keep the comparisons aligned. The three interactive evaluation files contain the same 2,800 unique request IDs. ACCESS trace hashes and terminal outcomes match its main-experiment pruning ledger and trace manifest. Serial and Policy request IDs, trace basenames, and outcomes match their respective manifests. This verification uses exported records. ACCESS's 3,016 interactions come from the archived audit of 2,800 traces; Serial and Policy totals come from per-request fields.

Table S10: Recorded interaction cost

Counter	Serial	Policy	ACCESS
User interactions, total	11,810	15,014	3,016
Logical calls, total	26,466	32,887	19,218
Logical calls, mean	9.45	11.75	6.86
Logical calls, median	7	7	5
Input tokens, total	91,075,986	135,317,418	72,198,669
Output tokens, total	1,505,248	2,017,583	3,559,259
Total tokens, mean	33,064.73	49,048.21	27,056.40
Total tokens, median	12,641.50	14,718.50	19,408.00

Note. **ACCESS uses fewer interactions and lower mean frontend tokens.** Recorded frontend counters for the shared population, including requester calls. SQL-generation tokens and unrecorded transport retries are outside this accounting.

Table S10 shows lower mean frontend tokens for ACCESS, although its median total tokens and mean output tokens are higher than those of Serial and Policy. The averages include refusals and reflect each workflow's terminal outcomes. Tables S17 and S18 test whether the saving persists among handoffs and per successful answer. Direct has no clarification frontend and is excluded from this cost comparison.

The field `logical_calls` counts recorded logical calls; transport retries can generate additional provider attempts. Section S10 separates logical reuse from recorded actual usage for the ablations, and Section S14 reports SQL-generation usage. The main-run counters cover recorded frontend effort, not a complete provider bill.

Configuration records identify the evaluated workflows. DeepSeek Flash is the display name in the main paper; the recorded API identifier is `deepseek-flash`. ACCESS's main outcomes are indexed by the `v10_allcase_shard` trace manifest. The `v10_full` executable snapshot records temperature zero, a shared recorder for decomposition, refinement, review, and requester calls, 30 turns, an 80-attempt shared budget, and `requester_protocol_v6_one_stage_gold`. The trace manifest identifies selected outcomes; the executable snapshot identifies implementation and settings. Section S10 specifies the frozen module workspace and the sequential variant's state-handoff repair.

Serial and Policy manifests record the same model alias and requester protocol, temperature zero, a 512-token response cap, 30 turns, and 80 calls. They also record shared runtime hashes and the original and adapted prompt hashes. Model identity is recorded at provider-alias granularity; source snapshots and selected traces are identified separately.

Identity checks link candidates to common scoring inputs. Request IDs and SQL hashes join selected candidates to effective policies and reference columns. The evaluator reconstructs authorization, tuple-wise execution agreement, order-aware SafeEX, SHR, BSS, and paired statistics. Frozen frontend sources provide the operation contracts and prompts; experiment manifests associate selected outcomes with their generation and recovery stages.

Persistent state connects clarification and deferred pruning. Following Section III and Fig. 2 of the main paper, the interaction state $X_t = (G_t, Q_t, O_t, B_t)$ maintains the requirement graph, question ledger, obligation pool, and deferred-blocking region. The graph records goals and necessary conditions; the ledger links questions and answers to their requirements; the obligation pool tracks contributions still needing resolution; and the blocking region stores proposed removals with their causes. These records let new answers update requirements and let access conflicts redirect later clarification. Dependency-aware review then checks how a removal affects surviving goals.

In Fig. 2 of the main paper, the label S denotes the schema; D, Π, and r denote the database, policy, and role. The diagram writes the state as $X = (G, Q, O, B)$ and omits the refinement-step subscripts.

State contracts separate semantic proposals from structural checks. The frozen implementation distinguishes output and modifier nodes, each with an identifier, parent lineage, business text, column bindings, and pending questions. Necessary dependencies use consumer-to-provider `requires` edges; annotation attachments use `applies_to`. A replacement maps an edited parent to one or more children. Existing attachments and contribution identities are transferred by code; new edges must involve a local replacement. Shared conditions are preserved by their identities and consumers, rather than by matching column names. The validator requires pending decisions to be routed, prior dependencies to be preserved or explicitly accounted for, and modifiers to retain an output consumer. These contracts check the submitted state; semantic proposals supply the requirement meanings and dependency classification.

Pruning review checks that retained goals preserve their meaning. The review compares requirements before and after pruning using their nodes and edges, permission facts, proposed deletions, the original request, actual answers, evidence quotations, and candidate handoff text. It checks for unsupported dependencies, missing intent, answer conflicts, and changed residual scope. Reported issues identify affected nodes and cite the supplied evidence. Acceptance commits the deletion; rejection rolls it back and returns unresolved. This review applies to proposed pruning; ordinary unpruned handoffs use the state and closure checks. Generated SQL then undergoes independent authorization and execution-result matching.

Rejecting a clarification menu records answer evidence for deferred review. The final refusal or handoff is determined after reviewing the affected requirements, and any generated SQL receives its own authorization check.

Failed local checks prevent an unvalidated handoff. In the frozen module workspace, contract validation permits up to two calls for shape/state repair. Transport exceptions are separate execution failures. For free-text answers, `answer_contribution_review` checks whether substantive answer content and necessary operands appear in the proposed binding claims. A negative coverage review rolls back the proposal and marks the attempt unresolved; it does not silently ask the user again. A final transition-audit failure also marks the attempt unresolved. Whole-attempt retries reported in Section S10 belong to the experimental runner, not to this internal scheduling step.

The requester gives each workflow access to the same intended task. Its private inputs are the reference question, intent fragments, Gold SQL, current question, and dialogue history. Only its answer is passed to the refinement agent. The construction procedure and requester use fragment-based intent descriptions, while ACCESS builds its own state from the visible request and received answers. All interactive pipelines use this requester protocol; five-model replay then varies SQL realization while keeping each pipeline's frontend decisions fixed.

Human review accepts 117 of 120 sampled rewrites. Two reviewers independently assessed one L0/L1 pair from each of 60 source questions, with 20 questions per dataset. Their consolidated assessment accepted 117/120 texts (97.5%). All 120 texts, including the three not accepted, remain in the reported experiments. This rate describes final acceptance of the sampled rewrites. The sample was selected after score analysis, and its paired texts represent 60 source questions; inter-reviewer agreement is not reported.

Human review confirms all 80 references and the negative cohort. Two human reviewers independently assessed all 80 references under the evaluation contract: authorization, preservation of required result tuples and duplicate multiplicity, and final ordering on the benchmark database. A final consolidated assessment accepted all 80 references (100%). Two human reviewers also independently examined every one of the 1,318 requests in Z; both found no additional legal service satisfying the original requirements.

S10 Ablation Protocol and Recovery

The three variants remove different workflow capabilities. *w/o clarification* suppresses user or simulator interaction but retains internal analysis and permission-based pruning. *w/o selective pruning* retains clarification and permissions, but returns a whole-request refusal when a blocked contribution requires removal. *w/o coordination* completes clarification before pruning and does not reopen questions during the pruning phase. Every variant retains policy access. Full ACCESS and the sequential variant both retain persistent state, so their comparison tests when updated state can guide further questions. The main Serial baseline differs: it introduces policy only at SQL generation.

Every variant is evaluated on the same requests. Each variant uses the identical 2,800 request IDs, including refusals, and generates SQL only for terminal handoffs. Selected trace hashes are checked against the selection manifest and retry audit for request identity, outcome, policy, and module flags. Full ACCESS reuses the DeepSeek Flash main run so its candidates and scores match the main comparison. Variant outcomes follow the recovery protocol below. Their SQL-generation settings are `deepseek-flash`, temperature zero, a 4,096-token cap, disabled thinking, no demonstrations, and no policy in the SQL prompt. The provider alias is the recorded model identifier.

Selection uses the first completed refinement, before SQL scoring. For each request, attempts are ordered and the first genuine `deliver` or `refuse` outcome is selected. Here, a genuine terminal outcome means a recorded handoff or refusal rather than an unresolved attempt or system error. Selection does not use SQL correctness or Gold scores, and failed attempts remain in the cost accounting. Recovery can repeat inputs or add validator-error feedback to a contract-repair prompt. Such feedback contains the visible request, prior response, validator diagnostics, and confirmed user choices, excluding Gold and SQL-evaluation feedback. Later recovery of the sequential variant permits additional contract-repair attempts under unchanged validators.

All 2,800 sequential-variant requests use the corrected state-handoff implementation. Its superseded implementation is archived separately and excluded from the reported scores and effort. The effort totals include 20 later terminal reruns caused by interrupted scheduling; these leave the earlier selected outcomes unchanged. Six interrupted calls have no usage record and are reported as missing cost.

Table S11: Ablation completion and cost

Quantity	w/o clarification	w/o selective pruning	w/o coordination
Final deliveries / refusals	1,806 / 994	1,294 / 1,506	1,478 / 1,322
Initial terminal outcomes	2,631	2,649	2,535
Initial completion (%)	93.96	94.61	90.54
Recorded attempts	3,194	3,232	3,549
Selected traces using feedback	2	5	2,440
All-attempt logical calls	10,900	16,069	30,783
All-attempt tokens	47,481,243	61,366,269	137,127,210
Tokens per fixed request	16,957.59	21,916.52	48,974.00
Interrupted calls, usage unknown	0	0	6

Note. **All variants reach terminal outcomes, with different recovery effort.** Completion and recorded frontend effort. Initial completion is the number of requests reaching a genuine terminal outcome on their earliest recorded attempt, divided by 2,800. Feedback counts selected traces supplied with validator diagnostics for contract repair, including first attempts made after feedback tooling was enabled. It differs from counts of user interactions and whole-attempt retries.

Table S11 shows final completion of all 2,800 requests per variant, compared with 90.54–94.61% on the earliest attempts. The sequential variant uses the most recorded recovery effort. Tokens include failed attempts and available interrupted-call usage; SQL generation and missing usage are excluded. Full ACCESS's historical counters in Section S9 have different retry coverage, so the cost tables retain the two accounting scopes separately.

Actual usage excludes reused responses. A trace's `cost` includes logical reuse, whereas `actual_cost` excludes responses reused without another model call. Rechecking 149 selected traces (two w/o selective pruning and 147 sequential) matches their frozen hashes and accounts for every difference using `reused_from` and recorded usage. The following table uses actual usage for selected attempts and the same accounting for all attempts. The additional percentage is (all-attempt tokens/selected-attempt tokens -1) $\times 100$; it includes failed attempts and recorded post-selection reruns, not solely retry overhead.

Table S12: Ablation token usage

	w/o clarification	w/o selective pruning	w/o coordination
Selected actual tokens (M)	40.078	51.784	91.495
All-attempt tokens (M)	47.481	61.366	137.127
Additional effort (%)	18.47	18.50	49.87

Note. **Counting all attempts reveals additional recorded effort.** Selected and all-attempt frontend tokens. M denotes millions. All-attempt totals exclude missing usage and SQL generation.

Table S12 shows that counting all attempts adds 18.47%, 18.50%, and 49.87% to selected-attempt token totals for w/o clarification, w/o selective pruning, and w/o coordination, respectively. The cost of reaching a terminal result is therefore especially important for the sequential workflow. Totals are recomputed from the 2,800 request-level attempt audits per variant. The sequential total includes the 20 later terminal reruns and excludes the superseded implementation; six interrupted calls have missing usage. These are recorded token totals, with SQL, manifests, attempt audits, and source hashes retained in the evidence archive.

S11 Selective Pruning and Coordination Preserve Partial Service

The ablations use the main paper's order-aware SafeEX and harmonic BSS on the same output-evaluation cohort E (1,482 requests) and recoverable-denied subset P (80 requests). Each request receives the fraction of target columns jointly matching complete result tuples, duplicate counts, and final Gold ordering, after whole-query authorization. Refusals remain in the denominators. As in the main table, extra authorized columns do not reduce SafeEX.

Reference IDs, policies, target columns, and selected Gold alternatives match the main evaluation. Native checks reconcile inconclusive candidate-authorization evidence; all 240 recoverable-denied request–variant records have resolved scores. The order-aware evaluator also performs 320 nontrivial candidate checks across 43 source clusters. The three variants contribute 4,446 recovery records and 4,194 handling records, paired with the unchanged full-ACCESS records. Rescoring keeps candidate SQL and output targets fixed.

Table S13: Ablation results

Variant	BSS	SafeEX	SHR	SafeEX$_A$	SafeEX$_P$	SafeEX$_{P_S}$
ACCESS (full)	67.23–67.95	54.28–55.23	88.27	54.60–55.60	48.75	55.32
w/o clarification	44.92–45.65	33.28–34.09	69.10	33.46–34.32	30.00	27.66
w/o selective pruning	66.87–67.88	51.74–52.96	94.49	53.77–55.05	16.25	6.38
w/o coordination	66.73–67.51	53.54–54.55	88.56	54.60–55.67	35.00	46.81

Note. **Full ACCESS recovers more denied-request output than each ablation.** Module-variant scores (%). P contains 80 recoverable denied requests; P_S contains 47 structurally supported partial references. All P and P_S scores are resolved. w/o coordination retains clarification and pruning in a fixed sequential schedule.

In these four DeepSeek arms, the P scores happen to be binary: positive credits recover every legal column. The metric nevertheless remains fractional-column SafeEX. Only A retains evidence uncertainty: full ACCESS has 14 empty-reference cases; w/o clarification has 12; w/o selective pruning has 17 empty-reference cases and one execution-pending case; w/o coordination has 15 empty-reference cases. These produce evidence bounds, not sampling intervals.

Tables S13 and S14 show 58 refusals and 22 safe deliveries on recoverable denied requests (P) without pruning, versus 15 and 59 for full ACCESS; complete recoveries rise from 13 to 39 with the full workflow. Selective pruning therefore preserves service that whole-request refusal often discards. Authorized partial SQL can earn credit in every

Table S14: Ablation handling and recovery

Variant	Z refusal	P refusal	P auth. SQL	P other	Full$_A$	Full$_P$	Full$_{P_S}$
ACCESS (full)	1,160	15	59	6	732	39	26
w/o clarification	912	8	46	26	434	24	13
w/o selective pruning	1,241	58	22	0	731	13	3
w/o coordination	1,166	18	54	8	735	28	22

Note. **Selective pruning preserves service that whole-request refusal discards.** Handling and complete-recovery counts. $Z/P/P_S$ contain 1,318/80/47 requests. Both refusal columns count whole-request refusals; authorized SQL denotes permission compliance. The three P handling columns partition all 80 requests. Full entries are counts of confirmed complete recoveries, not percentages.

arm, even without an internal pruning event; the higher score reflects recovered output rather than declarations of pruning.

Table S13 gives ACCESS the highest confirmed BSS endpoint, 67.23%, versus 66.87% for w/o selective pruning and 66.73% for w/o coordination, with overlapping evidence ranges. Its SafeEX$_P$ is 48.75%, versus 30.00%, 16.25%, and 35.00% for the three variants. These component scores locate the full workflow's clearest advantage in recovering authorized output from restricted requests; Section S12 evaluates the corresponding paired differences. In particular, the coordinated workflow recovers 39 complete references on P, compared with 28 for the sequential workflow, giving the 13.75-percentage-point gain reported in Section V-B of the main paper. The shared state carries answers, dependencies, and removal evidence through both workflows; coordination permits these updates to affect subsequent questions and pruning decisions.

S12 Adjusted Partial-Recovery Gains Support the Full Workflow

Each contrast is full ACCESS minus one variant, in percentage points. We use 10,000 paired, dataset-stratified source-question bootstrap replicates with seed 20261106. Overall and benchmark-allowed analyses use 476 clusters; recoverable denied requests use 28, and the structural subset uses 16. Each replicate recomputes the eligible denominators and forms BSS from that replicate's SafeEX and SHR. For inconclusive scores, subtracting opposite evidence endpoints widens the difference. The intervals describe source-question sampling with selected workflows and references held fixed.

Ordinary intervals use the 2.5% and 97.5% percentiles. Family-adjusted intervals use tails $0.05/(2 \times 3)$ and $1 - 0.05/(2 \times 3)$, applying a Bonferroni correction to the three variant comparisons within each metric and population. Separate metrics and populations define separate comparison families.

Table S15: Paired ablation contrasts

Population	Metric	Variant	Ordinary 95%	Family-adjusted
E	SafeEX	w/o clarification	[17.56, 24.50]	[16.92, 25.10]
E	SafeEX	w/o selective pruning	[-0.51, 5.24]	[-0.96, 5.61]
E	SafeEX	w/o coordination	[-1.96, 3.35]	[-2.30, 3.74]
$\mathcal{I}_{\text{den}}$	SHR	w/o clarification	[16.76, 21.61]	[16.21, 22.21]
$\mathcal{I}_{\text{den}}$	SHR	w/o selective pruning	[-8.07, -4.46]	[-8.51, -4.08]
$\mathcal{I}_{\text{den}}$	SHR	w/o coordination	[-1.57, 1.00]	[-1.86, 1.36]
$E, \mathcal{I}_{\text{den}}$	BSS	w/o clarification	[19.35, 25.17]	[18.86, 25.71]
$E, \mathcal{I}_{\text{den}}$	BSS	w/o selective pruning	[-2.18, 2.61]	[-2.56, 2.96]
$E, \mathcal{I}_{\text{den}}$	BSS	w/o coordination	[-1.65, 2.65]	[-1.92, 2.97]
A	SafeEX	w/o clarification	[17.51, 24.75]	[16.94, 25.36]
A	SafeEX	w/o selective pruning	[-2.07, 3.40]	[-2.47, 3.73]
A	SafeEX	w/o coordination	[-2.63, 2.52]	[-2.99, 2.88]
P	SafeEX	w/o clarification	[10.84, 27.16]	[8.86, 28.75]
P	SafeEX	w/o selective pruning	[20.00, 44.74]	[17.28, 47.50]

Population	Metric	Variant	Ordinary 95%	Family-adjusted
P	SafeEX	w/o coordination	[3.90, 24.10]	[2.47, 26.32]
P_S	SafeEX	w/o clarification	[16.67, 38.64]	[14.58, 41.30]
P_S	SafeEX	w/o selective pruning	[30.43, 66.67]	[26.09, 70.21]
P_S	SafeEX	w/o coordination	[-2.13, 21.74]	[-4.26, 25.00]

Note. **All three adjusted gains on recoverable denied requests are positive.** Paired cluster intervals for full ACCESS minus each variant (pp); P denotes the 80 recoverable denied requests. Sampling intervals are widened for unresolved evidence.

Table S15 shows that full ACCESS improves overall SafeEX, SHR, and BSS over w/o clarification under both interval conventions, and improves P recovery over all three variants after adjustment. On P_S, its adjusted recovery gains are positive against w/o clarification and w/o selective pruning. The adjusted BSS intervals against w/o selective pruning and w/o coordination are $[-2.56, 2.96]$ and $[-1.92, 2.97]$, respectively. The partial-recovery comparisons therefore identify the benefit more precisely than these aggregate contrasts. As reported in Section V-B, the adjusted coordination gain on P is $[2.47, 26.32]$ percentage points; on the structural subset P_S, the corresponding interval is $[-4.26, 25.00]$. The subset-specific intervals describe their respective populations.

S13 Exact Output Matching Preserves the Partial-Service Gain

For each request, let m_i be the candidate output-column count and L_i the fixed target-column set. Exact adds $m_i = |L_i|$ to the full-reference condition $c_i = 1$. All candidate columns must therefore participate in the injective reference match, retaining complete result tuples, duplicate multiplicity, and Gold final ordering. The same whole-SQL authorization gate applies.

Native result widths are matched by the exact dataset/database/SQL key. All 2,879 confirmed full recoveries across the four DeepSeek arms have resolved native width. The diagnostic reuses the latest ordered scores and changes no generated SQL.

Table S16: Exact output recovery

Pipeline	Cohort	N	Full count	Exact count	Extra
Direct	E	1482	456–470	384–396	72
Direct	A	1402	450–464	382–394	68
Direct	P	80	6	2	4
Serial	E	1482	815–852	768–801	47
Serial	A	1402	811–848	764–797	47
Serial	P	80	4	4	0
Policy	E	1482	837–880	794–835	43
Policy	A	1402	824–867	784–825	40
Policy	P	80	13	10	3
ACCESS	E	1482	771–785	746–760	25
ACCESS	A	1402	732–746	708–722	24
ACCESS	P	80	39	38	1

Note. **The partial-service gain survives exact output matching.** DeepSeek exact outputs. Counts use fixed cohorts; ranges reflect unresolved score evidence. Extra counts confirmed full recoveries with additional columns; unresolved records are excluded from that count.

Table S16 shows that ACCESS retains 38 of 39 complete positive-denied recoveries under exact columns; the corresponding counts for Direct, Serial, and Policy are 2/6, 4/4, and 10/13. Across E, additional-column cases account for 25/771 ACCESS full recoveries and 43/837 Policy full recoveries. The partial-service ranking persists under the exact-output criterion.

S14 Frontend Token Savings Persist per Handoff and Useful Answer

The cost comparison covers ACCESS, Serial, and Policy, each on the same 2,800 requests. Recorded frontend usage includes refinement and requester calls; Direct has no clarification frontend. The following means group requests by frontend outcome. A handoff proceeds to SQL generation, where it may still produce a refusal or failure. A dash denotes an empty outcome group.

Table S17: Cost by terminal outcome

Pipeline	Outcome	Requests	Mean tokens	Median tokens
ACCESS	deliver	1487	29.62	19.27
ACCESS	refuse	1313	24.15	19.46
ACCESS	all	2800	27.06	19.41
Serial	deliver	2800	33.06	12.64
Serial	refuse	0	–	–
Serial	all	2800	33.06	12.64
Policy	deliver	2381	48.89	14.13
Policy	refuse	419	49.94	30.07
Policy	all	2800	49.05	14.72

Note. **Lower frontend effort also holds among handoffs.** Frontend effort by terminal outcome. Tokens are in thousands.

Table S17 shows ACCESS using 29.62k frontend tokens per handoff, compared with 33.06k for Serial and 48.89k for Policy. The lower mean cost per handoff shows that the saving also occurs among requests passed to SQL generation.

Successful-service cost includes failed and refused requests. For complete recovery, divide all 2,800 frontend token totals by the number of confirmed Full_E requests. For positive partial service, divide the frontend tokens for all 80 P requests by the number with $c_i > 0$. Failed and refused requests remain in the relevant effort pool. These ratios measure the effort spent per successful result rather than the mean cost among successful sessions.

Table S18: Cost per successful service

Pipeline	Full_E count	Tokens/Full_E	P positive	P tokens/positive
ACCESS	771	98.26	39	87.27
Serial	815	113.60	6	676.52
Policy	837	164.08	14	421.06

Note. **ACCESS spends fewer frontend tokens per successful service.** Effort per confirmed success. Tokens are in thousands.

Table S18 gives ACCESS the lowest frontend cost per complete recovery (98.26k tokens) and per positive recovery on P (87.27k), charging recorded frontend tokens for failed and refused requests to the same cost.

Frontend and SQL-generation costs are accounted for separately. Provider records contain 9.647M SQL-generation tokens for Serial (2,801 responses for 2,800 final requests, including one additional response) and 7.026M for Policy (2,381 responses). Adding the selected frontend counters gives 102.228M and 144.361M recorded tokens, respectively. These totals cover the recorded calls; historical frontend transport retries and end-to-end latency were not logged.

The sequential module variant records 4.682M SQL tokens for all 1,478 deliveries. Added to 137.127M retry-inclusive frontend tokens, this gives 141.809M recorded tokens across both stages, or 50.646k per fixed request. Six interrupted frontend calls have no usage. Main ACCESS retains its SQL outputs without the corresponding provider usage log; w/o clarification and w/o selective pruning archives retain 78 and 84 newly generated responses alongside reused candidates. Their SQL-stage coverage is therefore reported separately from the complete frontend accounting in S10. The separate Direct run covers a larger generation population and is not pooled into this 2,800-request cost table.

Table VI of the main paper uses the common selected-frontend accounting for all three interactive pipelines. Reuse-aware all-attempt costs and SQL-response totals retain their distinct coverage, and recorded SQL execution seconds remain separate from inference latency.

S15 BSS Gains Survive Paired Testing and Weight Checks

BSS is the harmonic mean throughout the main paper and supplement. Each of 10,000 paired, data-source-stratified cluster replicates first computes SafeEX and SHR on their eligible request populations and then evaluates $\mathcal{H}(\text{SafeEX}, \text{SHR})$. Bounds propagate monotonically through $\mathcal{H}$. The table reports the lower-tail percentile of the lower difference and the upper-tail percentile of the upper difference. Ordinary tails are 0.025 and 0.975; family-adjusted tails are $0.05/(2 \times 15)$ and $1 - 0.05/(2 \times 15)$.

Table S19: Paired BSS gains

Model	Versus	Ordinary 95%	Family-adjusted
DeepSeek Flash	Direct	[19.92, 27.22]	[18.47, 28.48]
DeepSeek Flash	Serial	[4.81, 12.50]	[3.32, 13.76]
DeepSeek Flash	Policy	[23.39, 32.03]	[21.33, 33.73]
Qwen2.5-14B	Direct	[28.84, 36.77]	[27.27, 38.33]
Qwen2.5-14B	Serial	[18.61, 27.01]	[16.94, 28.64]
Qwen2.5-14B	Policy	[18.58, 26.84]	[16.93, 28.39]
Arctic R1-7B	Direct	[53.13, 60.85]	[51.38, 62.30]
Arctic R1-7B	Serial	[53.59, 61.33]	[51.73, 62.75]
Arctic R1-7B	Policy	[19.01, 27.59]	[17.44, 29.31]
Qwen2.5-7B	Direct	[31.60, 43.38]	[30.21, 44.88]
Qwen2.5-7B	Serial	[17.88, 29.63]	[16.46, 31.00]
Qwen2.5-7B	Policy	[23.62, 36.14]	[21.89, 37.85]
Qwen3.8-27B	Direct	[21.59, 31.50]	[19.87, 33.09]
Qwen3.8-27B	Serial	[2.78, 11.88]	[1.37, 13.09]
Qwen3.8-27B	Policy	[23.55, 33.82]	[21.83, 35.46]

Note. **All 15 adjusted BSS gains remain positive.** Paired BSS improvements, ACCESS minus each main comparator (pp).

Table S19 shows positive family-adjusted lower endpoints for all 15 BSS gains. ACCESS's combined-service advantage therefore survives paired sampling uncertainty and multiple-comparison correction across all five models. The computation preserves the same source clusters, sampled multiplicities, candidates, and component scores as the accompanying SafeEX and SHR analyses.

Let $x = \text{SafeEX}$ and $y = \text{SHR}$. For recovery weight $\alpha \in [0, 1]$ and $x + y > 0$, the weighted harmonic family is

$$\mathcal{H}_\alpha(x, y) = \frac{xy}{\alpha y + (1 - \alpha)x}, \quad 0 < \alpha < 1; \qquad \mathcal{H}_0(x, y) = y, \quad \mathcal{H}_1(x, y) = x.$$

For $0 < \alpha < 1$, the score is zero when either component is zero, including $x = y = 0$; BSS is $\mathcal{H}_{1/2}(x, y)$. Larger α places more emphasis on recovery. For each grid value, the table gives the minimum margin over all 15 comparisons after subtracting each comparator's upper evidence endpoint from ACCESS's lower endpoint. These are descriptive evidence margins, not bootstrap intervals.

Table S20: BSS sensitivity to weighting

α	Minimum margin
0.0	+10.52
0.1	+16.68
0.2	+15.21
0.3	+11.52
0.4	+8.30
0.5	+5.46
0.6	+2.97
0.7	+0.75
0.8	-1.22
0.9	-3.09
1.0	-7.09

Model	Disjoint B_{AZ} margin
DeepSeek Flash	+7.00
Qwen2.5-14B	+22.20
Arctic R1-7B	+22.90
Qwen2.5-7B	+19.51
Qwen3.8-27B	+4.38

Note. **The BSS lead persists through $\alpha = 0.7$ and without cohort overlap.** Conservative BSS margins under weighted emphasis and disjoint populations (pp).

Table S20 shows that the ranking holds across all five models through $\alpha = 0.7$ on the evaluated grid and also when P is removed from both components. At higher recovery emphasis, some comparator evidence ranges reach or exceed ACCESS. The main BSS uses equal harmonic emphasis, with both component scores reported alongside it.

S16 Planned-Population Completion

From 591 reviewed source questions, 476 have both a benchmark-allowed and a benchmark-denied role. Two roles and three specificity levels produce the 2,856 planned instances. The main paper reports the shared 2,800-request terminal cohort; the construction counts, nonterminal outcomes, and all-planned bounds are documented here.

The main comparison uses the same terminal request IDs for every pipeline. The completion ledger contains 2,856 planned requests: 1,487 handoffs, 1,313 refusals, 55 unresolved attempts, and one system error. The 2,800 handoff or refusal IDs define the shared main cohort. The other 56 IDs are excluded from every pipeline's paired score. This selection is based on ACCESS's frontend completion, so the analysis below evaluates how these exclusions could affect the main BSS ordering.

Table S21: Planned frontend completion

Source	Planned	Handoff	Refusal	Unresolved	System error
BIRD	1,026	515	501	9	1
Spider	1,128	605	506	17	0
LiveSQLBench	702	367	306	29	0
Total	2,856	1,487	1,313	55	1

Note. **Nonterminal attempts concentrate more in LiveSQLBench.** Terminal outcomes are handoff or refusal; unresolved and system-error attempts are not refusals. The table uses recorded frontend outcomes before SQL generation.

Table S21 gives nonterminal rates of 0.97%, 1.51%, and 4.13% for BIRD, Spider, and LiveSQLBench, respectively. L0/L1/L2 contain 18/21/17 nonterminal requests out of 952 each; benchmark-allowed and benchmark-denied groups contain 26 and 30 out of 1,428 each. The completion gap is therefore uneven across data sources. The ledger records outcomes, while finer causes such as unresolved dependencies, contract failures, or exhausted budgets require trace-level analysis.

Worst-case completion credit preserves the BSS ordering. We bound the possible effect of the 56 excluded requests without selecting targets from candidate outputs. Let $m_A = 26$ and $m_D = 30$ be their benchmark-allowed and benchmark-denied counts, and let $k \in \{0, \dots, m_D\}$ denote the number of recoverable denied requests among them. The all-planned denominators would be $|E| + m_A + k$ for SafeEX and $|\mathcal{I}_{\text{den}}| + m_D$ for SHR. Existing targets and per-request scores remain fixed. All excluded ACCESS attempts are unresolved or system errors, so they receive zero recovery and handling credit. For each comparator, we allow score one on every added recovery target and handling credit one on every added denied request.

Let U^-_{ACCESS} be ACCESS's existing recovery-score sum at lower endpoints and V_{ACCESS} its resolved handling-credit sum. For comparator j, let U^+_j and V_j be the corresponding upper recovery and resolved handling sums. Using the harmonic mean $\mathcal{H}$ from the main paper, let $\Delta_j(k)$ be ACCESS minus comparator j BSS on this expanded cohort. It obeys

$$\Delta_j(k) \;\geq \mathcal{H}\left(\frac{U^-_{\text{ACCESS}}}{|E| + m_A + k}, \frac{V_{\text{ACCESS}}}{|\mathcal{I}_{\text{den}}| + m_D}\right) - \mathcal{H}\left(\frac{U^+_j + m_A + k}{|E| + m_A + k}, \frac{V_j + m_D}{|\mathcal{I}_{\text{den}}| + m_D}\right).$$

As k increases, the ACCESS term decreases and the comparator term increases. The smallest difference therefore occurs at $k = m_D$, giving denominators of 1,538 for output fulfillment and 1,428 for safe handling. The bound assigns zero credit to excluded ACCESS outcomes and maximal credit to each comparator. It is deterministic and requires no assumption about random missingness; it is distinct from an observed all-planned score or a bootstrap interval.

Table S22 gives positive bounds for every model and comparator. The smallest margin is 2.23 points for Qwen3.8 versus Serial; DeepSeek's margin over Serial is at least 4.41 points. Thus even maximal comparator credit on the 56 excluded requests preserves the main BSS ordering under the fixed reference definition. The main tables continue to report the observed terminal cohort; exact all-planned scores and paired intervals would additionally need request-level references and comparator outcomes for the excluded IDs.

Table S22: Worst-case planned-population BSS bounds

Model	ACCESS lower	Direct upper	Serial upper	Policy upper	Min. gap
DeepSeek Flash	65.17	46.58	60.76	41.89	4.41
Qwen2.5-14B	56.88	28.73	38.31	38.45	18.43
Arctic R1-7B	57.40	6.30	6.08	38.39	19.01
Qwen2.5-7B	59.95	30.07	43.09	36.62	16.86
Qwen3.8-27B	63.46	43.69	61.23	40.07	2.23

Note. **Even maximal comparator credit preserves the BSS ordering.** All 2,856 planned requests are considered. The 56 excluded requests earn zero ACCESS credit; comparators receive maximal possible credit. Existing unresolved scores use ACCESS lower and comparator upper endpoints. Scores are percentages; the minimum gap is in percentage points across the three comparators. These are deterministic bounds.

S17 Constructing Three Specificity Levels

The construction procedure varies the explicitness of a source question before attaching role-conditioned benchmark records. It is an offline evaluation-data procedure, separate from ACCESS's online clarification and pruning. The generator receives the original natural-language question and database schema, including foreign keys and available column descriptions. Gold SQL, benchmark authorization labels, and role permissions are excluded from rewriting inputs. Gold SQL and role policies are subsequently preserved for evaluation.

Source spans define the controllable requirements. The generator identifies nonoverlapping verbatim spans for requested outputs and modifiers such as filters, grouping, ordering, and scope. A separate classification check distinguishes output goals from modifiers. Each span receives a semantic obligation stated in the context of the complete question; shared operators are normalized before editing. An inventory check identifies redundant introductory or population wording already covered by other output spans. Such context remains in the text but is excluded from the counted-fragment budget. Structural checks validate span positions, identities, coverage references, and nonoverlap.

One seeded permutation gives nested selection budgets. For N_{frag} counted fragments, a permutation with seed 42 fixes the order in which specifications are selected for generalization. At target retention $\rho \in \{0, \frac{1}{2}, 1\}$, the retained count is

$$N_{\text{keep}}(\rho, N_{\text{frag}}) = \operatorname*{arg\,min}_{k \in \{0,\ldots,N_{\text{frag}}\}} (|k - \rho N_{\text{frag}}|, k),$$

where k is a candidate retained-fragment count and minimization is lexicographic, preferring the smaller count on a distance tie. Thus L0 retains zero counted specifications in its plan, L1 retains $\lfloor N_{\text{frag}}/2 \rfloor$, and L2 retains all N_{frag}. Let F_ℓ^{hide} be the fragment identifiers selected for generalization at level $\ell \in \{0, 1, 2\}$. These sets satisfy $F_2^{\text{hide}} \subseteq F_1^{\text{hide}} \subseteq F_0^{\text{hide}}$; L2 returns the original question verbatim. These counts specify the construction plan; semantic preservation is checked in the review step below. The generalization planner may retain a selected fragment as context when hiding it would damage protected requirements or its specification remains determined elsewhere in the question. The implementation distinguishes the target budget, the rounded planned budget, and the fraction of fragments actually assigned a hiding action.

Generalization preserves a recoverable intended meaning. The default writer returns replacements for selected spans, and the program inserts them at the original positions. Coupled grammatical constructions may require clause-level rewriting; unsuccessful local edits admit a bounded whole-request fallback under the same checks. A hidden specification must become undecidable from the rewrite while the original requirement remains a possible clarification. A protected specification must remain required, rather than merely compatible with the rewrite. Review also checks the population, condition ownership, aggregation, conjunction, negation, and attachment of modifiers. The semantic-review call receives the source question, obligations, rewrite, and hidden/protected identifiers, but not the writer's hiding rationale. Failed or uncertain verdicts do not pass. A bounded repair may revise directions and regenerate the candidate; the recorded generation setting permits one repair.

The same question supports all role conditions. Candidate screening requires complete three-level records, passed generation review, and three distinct action-based effective-retention fractions; selection also uses anchor-retention and clarification-cue heuristics to form a review queue. The final export uses entries that pass the review contract. The review classifies each source requirement as preserved, recoverable through a question, silently lost, or contradicted. Each classification needs textual evidence, and each recoverable requirement needs a proposed

clarification question. Export joins the certified source question to the original RBAC records by database and normalized original text, retaining the Gold query, role policy, and benchmark label while replacing the request text. The resulting wording is identical across roles at a given level. Section S9 reports the independent human assessment of 120 exported texts, with final acceptance of 117/120.

The running example makes the three budgets concrete. The BIRD question in the construction example of Section IV-A of the main paper has three counted specifications: post-history count, last-edit date, and the post-title condition. Its exported variants are:

L2: Describe the post history counts and last edit date of the post title "What is the best introductory Bayesian statistics textbook?"

L1: Describe the post history details and a key timestamp of the post title "What is the best introductory Bayesian statistics textbook?"

L0: Describe summary information about the post's activity or history and a key timestamp-related attribute for posts identified by a defined selection criterion.

L1 generalizes the two output specifications and preserves the title condition, giving planned retention 1/3 after rounding. L0 additionally generalizes the title condition; L2 preserves the source text. The underlying database, Gold query, and per-role policy do not change.

Generation and review use separate recorded configurations. The three generation manifests record `gpt-5.1` for writing and `gpt-5.5` for review, seed 42, retention targets $[0, 0.5, 1]$, local replacement mode, and one repair. These calls construct evaluation data. ACCESS's frontend and the five SQL-generation models use their own configurations. The supplied implementation matches the recorded Spider and LiveSQLBench source fingerprints; BIRD records a different implementation version. Each dataset's prompts match its manifest. The cross-dataset entry point adds an instruction to copy source spans exactly.

Export checks preserve the intended pairing across levels and roles. Each level of the export pool contains 4,031 role records from 591 source questions, before the cohort selection in Section S16. Every L2 text equals its source question, and selected-fragment sets are nested and follow the L1 rounding rule. Gold SQL, roles, policies, benchmark labels, and other non-level fields remain identical across levels; wording is identical across roles at a given level.

Selected fragment IDs and rewritten texts remain available, allowing planned retention to be computed. The intermediate export omitted effective-retention metadata, so that field is null and no observed effective-retention distribution is reported. The supplied generation contracts pass all 27 existing unit tests with network access disabled. Together, these checks establish the structural pairing, while the human review in Section S9 assesses semantic preservation.